Zeke Sy

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Employment History

2025/10 – Present **Eyes Japan** *Full-time · Systems Engineer*

- **Sim-to-Real Team Lead:** Led a team of engineers developing sim-to-real transfer pipelines for robotic arms and quadcopters using Isaac Sim and Isaac Lab. Designed and managed the end-to-end pipeline from RL policy training in simulation through SITL and HITL validation to physical hardware deployment. Implemented ROS2-based control systems, real-time object tracking with onboard cameras, hand-eye calibration, and domain randomization strategies to close the sim-to-real gap. Delivered working demos under tight deadlines and drove key optimizations including control loop decoupling and ROS2 QoS tuning that resolved critical lag and oscillation issues on physical hardware.
- **Antibody Web Dashboard Lead:** Leading full-stack development of a web dashboard for antibody data visualization and analysis, overseeing architecture decisions, feature delivery, and cross-functional collaboration.
- **School Administrative System Maintenance:** Maintaining and improving a production school administrative system, handling bug fixes, feature updates, and system stability.
- **Blood Glucose Monitoring Platform (GlucoReview):** Owned full-stack development of a healthcare web platform enabling patients to schedule weekly home blood glucose tests and clinicians to track results and trends. Built with Next.js, Supabase, and PostgreSQL with role-based access control and HIPAA-aligned data handling practices.

2023/08 – 2025/09 **Eyes Japan** *Part-time · Software Developer*

- **Hearing and Balance Medical iOS App:** Led development of new features and improvements for a clinical-grade hearing and balance iOS application. Built pure-tone audiometry test screens with custom audio tone generation, white noise masking, and CSV audiogram export for clinical review using Unity and Swift.
- **Android Health and Fitness Application:** Contributed to front-end development of a Parkinson's disease monitoring app, building patient-facing screens and interactive data visualisation components for exercise and heart rate tracking using Kotlin and Jetpack Compose.
- **Weekly Report Automation System:** Designed and implemented an automated weekly staff progress report system in JavaScript, automating slide generation and integrating input forms for data capture and backend processing.

Employment History (continued)

2024/12 – 2025/02 **DeepAI** *Internship · AI Researcher*

- Conducted research into state-of-the-art image generative models including diffusion models and GANs, evaluating text-to-image and image-to-image architectures for real-world applicability.
- Evaluated and benchmarked generative models across quality, resource efficiency, and scalability metrics to assess production readiness.
- Analyzed existing SaaS platforms utilizing generative models, identifying potential improvements and opportunities for integration into applied pipelines.

2022/09 – 2022/10 **Technistock** *Internship · Beta Tester*

- Participated in early testing phases of new mobile application features, identifying and reporting usability issues, bugs, and performance bottlenecks.
- Developed and maintained automated test cases and reusable test flows using Katalon Studio, improving testing efficiency and reducing manual testing effort.
- Collaborated with the QA team and developers to provide actionable feedback that directly contributed to product improvements prior to release.

Education

2024 – 2025 **M.Sc.. Computer Science Honors Program, Aizu University**
Thesis title: *Multi-View CLIP Image-text Similarity Metric for evaluating Semantic Communication Systems*

2020 – 2024 **B.Sc. Computer Science, Aizu University**
Thesis title: *An Exploration of MLP and Grid Trade-off in Hybrid NeRF Models*.

Research Publications

Conference Proceedings

1 Z. R. Lim Sy, S. Luo, J. Flaherty, and T. C. Thang, "Towards green nerf: An exploration of energy influence factors in nerf models," in *2024 IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS)*, 2024, pp. 65–71. DOI: 10.1109/IoTaIS64014.2024.10799268.

Skills

Languages — Fluent reading, writing and speaking competencies for English; Upper-intermediate reading, writing and speaking competencies for Japanese (JLPT N2);

Coding — Python, Java, C, C#, TypeScript, JavaScript, Kotlin, SQL

AI / ML — Machine Learning, Deep Learning, Neural Networks, Computer Vision, Reinforcement Learning, Neural Radiance Fields (NeRF), Diffusion Models, GANs, Semantic Communication, Image Quality Assessment, PyTorch, TensorFlow, OpenCV, CLIP

Skills (continued)

Robotics	ROS2, Isaac Sim, Isaac Lab, Sim-to-Real Transfer, SITL / HITL Testing, Hand-Eye Calibration, Visual Servoing, TF Tree, Forward Kinematics, Domain Randomization
App Dev	Unity, iOS Development (Swift / Xcode), Android Development (Kotlin, Jetpack Compose), React Native, Digital Signal Processing, BLE
Web Dev	HTML, CSS, JavaScript, TypeScript, React, Next.js (App Router), Supabase (Row-Level Security, Edge Functions), PostgreSQL, Tailwind CSS
Tools	Git / GitHub, Docker, Supabase, Jira, Slack, Katalon Studio, Linux, VS Code
Misc.	Technical leadership, academic research, IEEE publication, consultation, LaTeX typesetting and publishing

Miscellaneous Experience

Awards and Achievements

2024-2025 **University of Aizu Honors Program Scholarship**

2020-2023 **MEXT Scholar**, Qualified and received MEXT Scholarship

Certification

2025 **JLPT**. Succesfully passed N2 exam.

2023 **JLPT**. Succesfully passed N3 exam.

2022 **TOEIC**. Scored 980 on the TOEIC exam